

Brendan Francis O'Connor

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Experience

Staff Security Engineer, Data Security

2024 - 2026

GitLab

As a Staff Security Engineer and one of two founding engineers on the Data Security team, I helped to find and eliminate security debt across GitLab products and infrastructure, and improve GitLab's product security posture. Among other projects: I reset our approach to database field encryption key management, saving (management estimate) \$3-5M of operating expenses incurred by taking the invalid-but-recommended approach; I remedied an ancient backup system, saving \$1M/y+ in cloud costs; I upgraded our password storage (and became the hashing SME); and I fixed a ten-year-old bug causing us to retain plaintext copies of highly-sensitive encrypted database values.

Chief Technical Officer / DSS

2010 -

Malice Afterthought, Inc.

Malice Afterthought provides information security consulting services. Some highlights of past work: I led an international nonprofit's security efforts, including both building a multi-regulated compliance program (FERPA, HIPAA, and FedRAMP) and implementing technical defensive tools; I taught at a Department of Defense information and network warfare (CNO) school; and I won and completed two DARPA Cyber Fast Track (CFT) contracts. Languages and tech stacks have included Ruby, Python, Perl, D3JS, UnityScript, Hadoop, Spark, and AWS, among many others.

Software Engineer, Security [Staff-Level]

2023 - 2024

Render

I guided major efforts to uplift Render's security and compliance posture, including: on day one, identifying and beginning remediation on a whole-of-company security and legal compliance issue (requiring cross-functional work between legal, technical, sales, and executive staff); and rebuilding several different engineering systems to enable least-privilege access across the corporation.

Staff Security Engineer

2022 - 2023

Cedar

As a Staff Security Engineer, I was responsible for engaging in high-leverage work to improve security across Cedar to protect patients, medical providers, and our employees; I also helped educate our engineers on secure programming practices, performed code reviews, and guided technical implementations of key client-required security features. Among other projects: I rebuilt Cedar's IAM infrastructure and secrets management systems; I compiled the first complete set of client contractual requirements for security; and I created a trusted baseline and deployment tracker for our container build systems.

Senior Security Advisor

2014 - 2016, 2019 - 2022

Leviathan Security Group

I worked as a Senior Security Advisor in Leviathan's Risk and Advisory Services group, where I built security programs, advised teams on how best to meet compliance goals, and conducted large-scale architecture and security ecosystem assessments on codebases with more than ten years' engineering effort, more than 20 million end users, or both. I built SOC 2 Type II-compliant security programs and defended them during audits, and advised management at dozens of companies on HIPAA, FERPA, HITRUST, CMS ARS, ISO 27001/2/18, and PCI compliance. I also co-wrote four whitepapers on forced data localization and one on

Kubernetes and container security.

Enterprise Security Platform Lead

2018 - 2019

RealSelf

As the head of the information security program at RealSelf, I directed the company's overall security, risk and compliance efforts. This involved close coordination with the RealSelf executive and engineering teams to align security priorities with our business goals and overall risk posture; I reported to the CTO and General Counsel. My work included developing and delivering security training, working with developers and SREs to design and implement security at all levels of our systems, building our ISO 27000-aligned information security management program, communicating with customers regarding security issues, leading security incident response efforts, working with our legal and privacy teams on vendor management, and managing our bug bounty program. I also worked on HIPAA, PCI, GDPR, and other regulatory issues to ensure that we were able to meet evolving obligations, and ultimately, that the trust our users placed in us was well-earned.

Selected Presentations

Security by Consent; or, Peel's Principles of Security Operations

October 18, 2016

SecTor - Toronto, ON, Canada

How to create and maintain a security operation within a larger organization that focuses on cooperation and consent, rather than coercion, based upon the 'policing by consent' model created in 1820s England.

Stalking a City for Fun and Frivolity

August 3, 2013

DEF CON - Las Vegas, NV

Distributed sensor network data acquisition, filtering, and visualization, with several security applications. This presentation addressed the architecture and collection aspects of the CreepyDOL system, as well as the implications for global surveillance states and engineering as a profession.

CreepyDOL: Cheap, Distributed Stalking

August 1, 2013

Black Hat USA - Las Vegas, NV

Distributed sensor network data acquisition, filtering, and visualization, with several security applications. This presentation addressed the architecture and collection aspects of the CreepyDOL system.

Education

Master of Science in Engineering in Computer Science

September 2005 - May 2009

Bachelor of Science in Computer Science

September 2004 - May 2008

The Johns Hopkins University

Certifications & Memberships

Fellow of Information Privacy (FIP)

June 6, 2023

Certified Information Privacy Technologist (CIPT)

May 10, 2023

Certified Information Privacy Professional U.S. Private Sector (CIPP/US)

December 14, 2015

International Association of Privacy Professionals

Certified Information Security Manager (CISM)

February 6, 2020

Certified Information Systems Auditor (CISA)

August 12, 2016

Information Systems Audit and Control Association

Certified Information Systems Security Professional (CISSP)

July 20, 2012

International Information Systems Security Certification Consortium (ISC(2))